

Metadata Reliance in Vision-Language Models for CT Lung Nodule Malignancy Prediction

Article Type: Original Research

Summary Statement: Structured prompt metadata, rather than computed tomography images alone, accounted for much of apparent vision-language-model performance in a retrospective lung nodule malignancy benchmark.

Key Points

- In 917 LUNA25 nodules, visual-only Gemini 3 Flash prompting achieved AUC 0.682–0.712; adding structured metadata raised the strongest condition to AUC 0.739, while a matched text-only control reached AUC 0.693.
- In a sparse-montage audit, Claude Opus 4.8 and Gemini 3.1 Pro rose from near-chance image-only performance to metadata-assisted AUCs of 0.730 and 0.729, respectively; GPT-5.5 image-only outputs were retained only as an interface-provenance check.
- The strongest metadata-assisted vision-language model remained below CT-only supervised three-dimensional deep learning, with AUC 0.739 versus 0.872 for STU-Net.

Abstract

Purpose: To determine whether general-purpose vision-language model (VLM) performance for CT lung nodule malignancy prediction reflects visual interpretation or prompt-supplied structured metadata. **Materials and Methods:** This retrospective benchmark used LUNA25 (917 annotations from 318 patients; 81 malignant) and LNDb external validation (814 rows; 74 malignant). Gemini 3 Flash Preview underwent six prompt/input conditions plus a matched text-only control. Hosted VLM inference was performed on April 30, 2026. Claude Opus 4.8 and Gemini 3.1 Pro were evaluated in a zero-shot sparse-montage audit; GPT-5.5 metadata-assisted and text-only rows were reported separately because image-only delivery failed in the tested interface. Comparators were seven supervised 3-D deep-learning (DL) models, Brock, and structured-metadata baselines; performance was assessed with ROC-AUC and paired DeLong testing. **Results:** Visual-only Gemini conditions reached AUC 0.682–0.712; adding structured metadata raised the 20-shot rich condition to AUC 0.739. The matched text-only control reached AUC 0.693, and F3 exceeded F0 by 0.047 AUC units (paired DeLong $P = .02$). In the sparse-montage audit, metadata increased AUC by 0.233 for Claude Opus 4.8 and 0.213 for Gemini 3.1 Pro; text-only controls recovered most metadata-assisted signal. The strongest VLM setting remained below supervised 3-D DL (best AUC, 0.872; $P < .001$). On LNDb, visual-only Gemini remained below all seven DL baselines. **Conclusion:** Prompt-supplied structured metadata accounted for substantial apparent VLM performance in this retrospective CT lung nodule benchmark. Text-only and metadata-control conditions are needed when evaluating multimodal VLMs with structured clinical descriptors.

Keywords: vision-language model, lung nodule, malignancy prediction, chest CT, metadata reliance, external validation

Abbreviations

AUC = area under the receiver operating characteristic curve; CT = computed tomography; DL = deep learning; ECE = expected calibration error; HU = Hounsfield unit; PR-AUC = precision–recall area under the curve; VLM = vision-language model.

1 Introduction

Lung cancer screening with low-dose CT reduces mortality in high-risk populations, but characterization of screen-detected pulmonary nodules remains difficult because benign and malignant nodules overlap in appearance (1, 2). Supervised 3-D deep-learning (DL) models trained directly on volumetric CT labels can perform well on this task (3), while general-purpose vision-language models (VLMs) and radiology foundation models are increasingly proposed as flexible clinical imaging assistants (4–6).

For volumetric CT, however, apparent VLM accuracy may combine image interpretation with signals supplied in the prompt. Recent medical VLM evaluations have emphasized that high aggregate accuracy can conceal shortcut behavior, robustness failures, and benchmark-interface artifacts (7–9). Lung nodule prompts often include structured descriptors such as size, margin, attenuation, smoking status, and location; these variables are themselves diagnostic predictors and overlap with established clinical rules such as Brock. Without text-only and metadata-aware controls, metadata-assisted VLM performance can be mistaken for visual CT reasoning.

We audited this issue for CT lung nodule malignancy prediction using LUNA25 and LNDb. We combined a deep Gemini 3 Flash Preview prompt/input ablation, a matched text-only control, a cross-family sparse-montage audit in Claude Opus 4.8, Gemini 3.1 Pro, and GPT-5.5, and comparisons with supervised 3-D DL, Brock, and structured-metadata baselines. We hypothesized that VLMs would show nontrivial signal but remain below task-specific 3-D DL, and that a substantial fraction of metadata-assisted VLM performance would be recoverable without images.

2 Materials and Methods

2.1 Study Design and Data

Because this retrospective computational benchmark used public, de-identified datasets, institutional review board approval and informed consent were not required (Figure 1). Reporting was aligned with CLAIM, STARD-AI, and TRIPOD-AI/LLM principles for AI diagnostic and prediction-model studies (10–13). The internal test set was the fixed patient-disjoint LUNA25 test split (917 annotations from 318 patients; 81 malignant, 836 benign) derived from NLST nodule-

centered CT crops (14, 15). Structured prompt fields included LUNA25 nodule descriptors and matched NLST demographic/smoking fields. Self-reported race was included in the primary Z3/F3 clinical text because it was present in the matched NLST descriptors; race-excluded tabular sensitivity analyses are reported separately. External validation used the DL-matched LNDb 814-row sheet (740 benign, 74 malignant; 439 unique findings) (16).

2.2 VLM Inputs and Controls

For the primary Gemini experiments, each $64 \times 128 \times 128$ CT crop was clipped to $[-1000, 1000]$ HU and encoded as a 20-fps MP4 video following the CT-to-video approach of Xu et al. (17). Gemini 3 Flash Preview was tested under six conditions: Z1/Z2/Z3 for zero-shot minimal, rich, and rich+clinical prompts, and F1/F2/F3 for the corresponding 20-shot conditions. F0 was a matched text-only control retaining the F3 prompt scaffold, exemplar labels/text, target clinical text, JSON schema, model identifier, and temperature, but removing all CT videos. Z0 was reserved for the zero-shot metadata-only condition in the cross-family sparse-montage audit, where no exemplar text or CT image was provided. Hosted VLM inference was performed on April 30, 2026. Calls used temperature 0 where supported and required a JSON response with a malignancy confidence score and one-sentence rationale; compact condition definitions and input routes are provided in the Supplement, and exact model identifiers, serialized prompt templates, software versions, and parsed outputs are provided in the anonymized review repository. Four additional F3 repeats formed the averaged F3A sensitivity estimate. Gemini 2.5 Flash, Gemini 2.5 Pro, Gemini 3.1 Pro Preview, and MedGemma 1.5-4B were evaluated under F3 for context.

For the cross-family audit, Claude Opus 4.8, Gemini 3.1 Pro Preview, and GPT-5.5 were evaluated through the same hosted model interface using Z2 image-only, Z3 image+metadata, and Z0 metadata-only prompts. Image rows used a standardized axial montage of frames 9, 29, and 49 from each 64-frame clip, matching the sparse frame positions observed in the hosted Gemini video sampling probe. The same montage audit was also run for Gemini 3 Flash Preview as a bridge to the primary video-input ablation. Matching frame indices does not make the montage pixel-token equivalent to native video ingestion, because the three frames are concatenated into a single image and may undergo different resizing, layout parsing, and modality-specific preprocessing. This audit was designed to test whether metadata reliance generalized across hosted VLM interfaces

under a portable sparse visual input, not to rank vendor-native image or video ingestion. GPT-5.5 image-only responses were constant at 0.5 with no-image rationales and were retained only as an interface-provenance check.

2.3 Comparators and Statistical Analysis

Seven supervised DL baselines were trained/evaluated on the same LUNA25 split: ResNet-18/50 (18), DenseNet-121 (19), EfficientNet-B0 (20), STU-Net (21), ViT-Base (22), and SwinUNETR (23). We also evaluated exploratory 20-shot Med3D ResNet18+TTT variants (24), the parsimonious Brock model (25) on 701 cases with complete mappable metadata, and classical structured-metadata models trained on the LUNA25 training split. The primary endpoint was ROC-AUC. The primary contrasts were F3 versus F0 and F3 versus the strongest supervised CT-only baseline. Pairwise AUC comparisons used paired DeLong tests (26); 95% CIs used stratified bootstrap resampling, with patient-cluster and LNDb finding-cluster sensitivity analyses. Other DeLong tests, calibration, PR-AUC, decision-curve net benefit (27), subgroup analyses, and rationale audits were exploratory and were not adjusted for multiplicity. Analyses were performed in Python 3 using scikit-learn, SciPy, NumPy, and pandas, with matplotlib for figures; a paired DeLong implementation was used for AUC comparisons, and two-sided $P < .05$ denoted statistical significance. Pinned package versions, per-sample outputs, and scripts are released.

3 Results

3.1 Gemini Prompt/Input Ablation

The LUNA25 test set included 917 nodules; mean age was 62.6 years, 182/318 patients (57.2%) were male, and 150/318 were current smokers. Visual-only Gemini conditions clustered at AUC 0.682–0.712 (Table 1; Figure 2). Adding structured clinical text raised the strongest setting, Gemini F3, to AUC 0.739 (95% CI: 0.677, 0.798), while the matched F0 text-only control reached AUC 0.693 (95% CI: 0.634, 0.746). F3 exceeded F0 by 0.047 AUC units (paired DeLong $P = .02$), but F0 correlated more strongly with Brock probability ($\rho = 0.723$) than image-only F2 ($\rho = 0.147$), indicating substantial structured-text signal. Cross-generation F3 runs were similar for Gemini 3 Flash Preview and Gemini 3.1 Pro Preview (AUC 0.739 vs 0.736), lower for Gemini 2.5 models

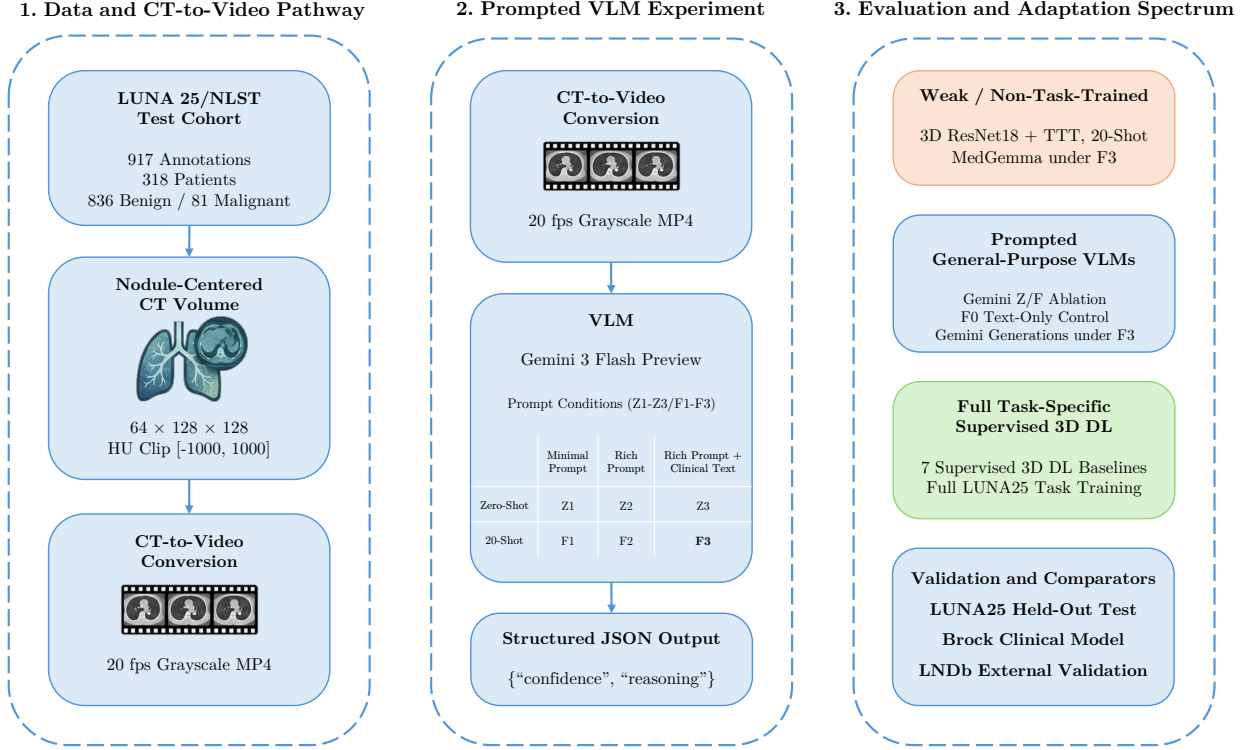


Figure 1: Study design and evaluation pipeline. Prompt/input conditions are labelled Z1–Z3 for zero-shot prompts (minimal, rich, rich+clinical), F1–F3 for the corresponding 20-shot conditions, and F0 for the matched F3 text-only control. Z0 denotes the separate zero-shot metadata-only control used in the cross-family sparse-montage audit.

84 (AUC 0.682–0.710), and lower for MedGemma 1.5-4B (AUC 0.594).

Table 1: Gemini 3 Flash Preview prompt/input ablation on LUNA25.

Condition	CT video	Clinical text	Shots	AUC	95% CI
Z1 minimal	Yes	No	0	0.702	0.639–0.762
Z2 rich	Yes	No	0	0.682	0.615–0.746
Z3 rich+clinical	Yes	Yes	0	0.730	0.671–0.783
F1 minimal	Yes	No	20	0.712	0.661–0.767
F2 rich	Yes	No	20	0.699	0.638–0.756
F3 rich+clinical	Yes	Yes	20	0.739	0.677–0.798
F0 text-only	No	Yes	20	0.693	0.634–0.746

Z denotes zero-shot and F denotes 20-shot/few-shot. F0 retained the F3 text scaffold and exemplar labels/text but removed all CT videos. F3 versus F0 paired DeLong $P = .02$.

85 3.2 Cross-Family Metadata Audit

86 Under the standardized three-slice montage, image-only performance was near chance for the Gem-
 87 ini 3 Flash bridge (AUC 0.508), Claude Opus 4.8 (AUC 0.497), and Gemini 3.1 Pro (AUC 0.516),

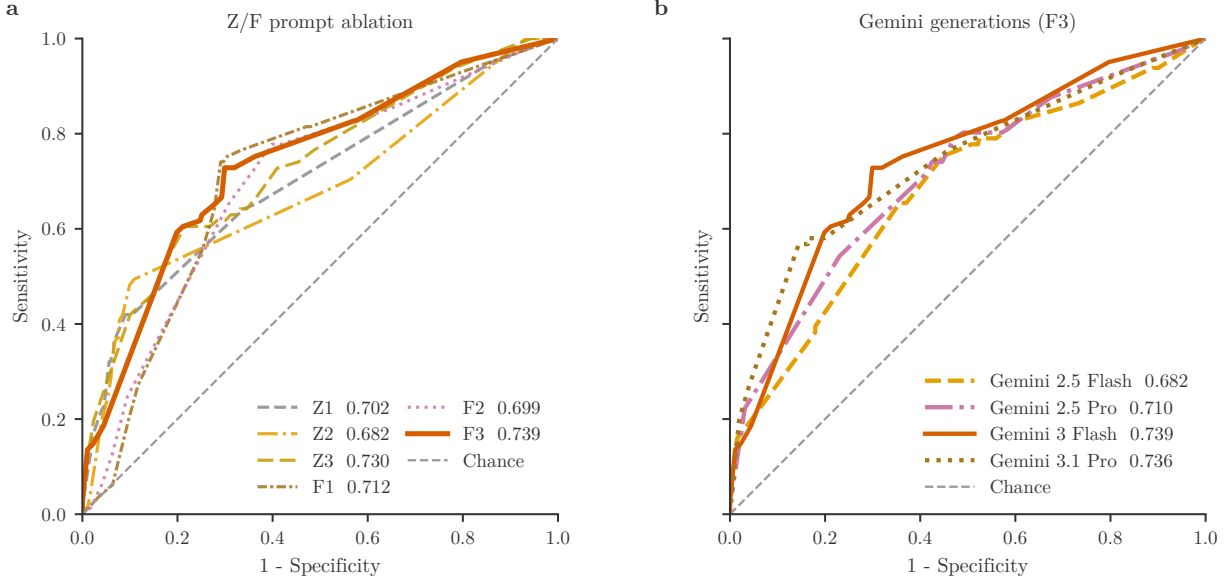


Figure 2: Receiver operating characteristic curves for the Gemini 3 Flash Preview Z/F prompt ablation and the Gemini-generation F3 comparison.

whereas adding structured metadata raised these rows to AUC 0.72–0.73 (Table 2). Text-only controls recovered most of the metadata-assisted signal (AUC 0.721, 0.719, and 0.713, respectively). Brock correlations rose from near zero in image-only rows to 0.77–0.84 after metadata was added. GPT-5.5 image-plus-text and text-only rows were similar (AUC 0.712 and 0.703), but its image-only row was not interpreted because the interface did not appear to deliver the image.

Table 2: Cross-family zero-shot sparse-montage audit on LUNA25.

Model	n_{pair}	Z2	Z3	Lift	Z0	ρ_{Brock}	Z2/Z3
Gemini 3 Flash video reference	917	0.682	0.730	+0.047	–	0.18/0.64	
Gemini 3 Flash montage	917	0.508	0.722	+0.214	0.721	–0.01/0.77	
Claude Opus 4.8	917	0.497	0.730	+0.233	0.719	–0.01/0.84	
Gemini 3.1 Pro	917	0.516	0.729	+0.213	0.713	0.06/0.79	
GPT-5.5	917	0.500 [‡]	0.712	NA [‡]	0.703	NA/0.82	

Z2=rich image-only, Z3=rich image+metadata, Z0=metadata-only. The video-reference row uses the original Gemini video-input ablation; other rows use the standardized single-image three-slice montage. The video-reference and montage rows share the nominal sampled frame positions but differ in ingestion path and preprocessing. Paired Z2-versus-Z3 DeLong P values for interpretable image-delivery rows were .23, <.001, <.001, and <.001, respectively. [‡]GPT-5.5 image-only responses were constant 0.5 with no-image rationales, indicating failed image delivery in the tested interface; this row was not included in image-only, lift, or paired Z2-versus-Z3 interpretation.

3.3 Comparison with Supervised DL, Brock, and External Validation

Full-data supervised 3-D DL remained strongest (Table 3; Figure 3). Gemini F3 received both CT video and structured metadata, while DL baselines used CT only; nevertheless, Gemini F3

was below six of seven DL baselines. The paired AUC gap was -0.133 versus STU-Net (AUC 0.872; DeLong $P < .001$) and -0.124 versus EfficientNet-B0 (AUC 0.863; DeLong $P < .001$). On the 701 Brock-computable cases, Brock reached AUC 0.774 and Gemini F3 reached AUC 0.750 (DeLong $P = .40$). A structured-only gradient-boosting model reached AUC 0.710, overlapping the visual-only Gemini plateau and exceeding F0 text-only prompting.

Decision-curve and calibration analyses supported the discrimination results. STU-Net had higher PR-AUC than Gemini F3 (0.520 vs 0.252), and raw-score net benefit at a 10% threshold was $+0.031$ for STU-Net and $+0.003$ for Gemini F3 (Supplement). Gemini F3 was poorly calibrated (Brier 0.222, ECE 0.270), whereas STU-Net combined higher AUC with lower Brier score (0.090). At threshold 0.5, Gemini F3 misclassified 280/917 cases; representative discordant cases are shown in Figure 4.

On LNDb, where comparable structured metadata were unavailable, visual-only Gemini prompting remained below all seven DL baselines (Table 4). Gemini F2 reached AUC 0.654 and Gemini Z2 reached AUC 0.642; EfficientNet-B0 reached AUC 0.815 and exceeded Gemini F2 by 0.161 AUC units (paired DeLong $P < .001$). The qualitative ranking was unchanged under finding-cluster bootstrap.

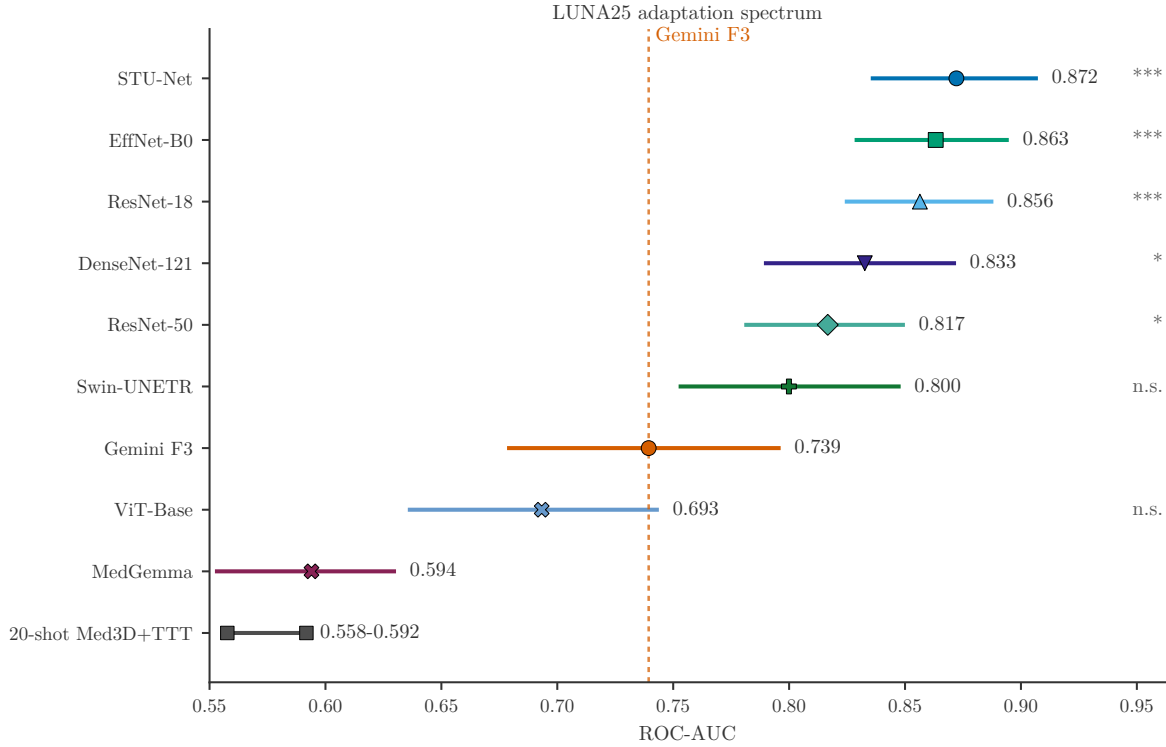
4 Discussion

This benchmark shows that general-purpose VLMs can produce moderate malignancy scores for CT lung nodules, but that their strongest apparent performance was materially influenced by prompt-

Table 3: Performance spectrum on LUNA25.

Method	Type	AUC	95% CI
STU-Net	DL	0.872	0.833–0.906
EfficientNet-B0	DL	0.863	0.827–0.898
ResNet-18	DL	0.856	0.823–0.887
DenseNet-121	DL	0.833	0.789–0.874
ResNet-50	DL	0.817	0.782–0.848
Swin-UNETR	DL	0.800	0.752–0.846
Gemini F3	GP-VLM	0.739	0.677–0.798
ViT-Base	DL	0.693	0.635–0.744
MedGemma 1.5-4B (F3)	Spec. VLM	0.594	0.555–0.630
20-shot Med3D ResNet18+TTT	Weak DL	0.558–0.592	observed range

GP-VLM=general-purpose VLM; Spec. VLM=specialty medical-imaging VLM. DL baselines used CT volumes only; Gemini F3 additionally received structured metadata.



Error bars: 95% bootstrap CI; 20-shot Med3D+TTT: observed range. DL markers: paired DeLong vs Gemini F3.

Figure 3: Performance spectrum on LUNA25. Points and error bars show AUCs with 95% bootstrap CIs for supervised DL baselines, Gemini F3, MedGemma 1.5-4B, and weak-adaptation Med3D ResNet18+TTT variants. Stars denote paired DeLong tests versus Gemini F3: $*P < .05$, $**P < .01$, $***P < .001$; n.s.=not significant.

Table 4: External validation on LNDb using the DL-matched 814-row evaluation sheet.

Method	Int. AUC	LNDb AUC	Row 95% CI	Cluster 95% CI
EfficientNet-B0	0.863	0.815	0.761–0.863	0.760–0.866
ResNet-18	0.856	0.812	0.759–0.864	0.750–0.865
DenseNet-121	0.833	0.807	0.754–0.858	0.746–0.861
STU-Net	0.872	0.803	0.742–0.863	0.740–0.862
ResNet-50	0.817	0.762	0.706–0.816	0.701–0.821
ViT-Base	0.693	0.683	0.624–0.743	0.612–0.750
Swin-UNETR	0.800	0.666	0.596–0.728	0.592–0.737
Gemini F2	0.699	0.654	0.592–0.715	0.587–0.724
Gemini Z2	0.682	0.642	0.575–0.704	0.569–0.706

LNDb contained 740 benign and 74 malignant rows representing 439 unique findings. VLM external runs used visual-only Z2/F2 prompts because comparable structured clinical fields were unavailable.

115 supplied structured metadata. Visual-only Gemini conditions plateaued near AUC 0.70, whereas
116 adding structured descriptors raised AUC to 0.739. A matched text-only control recovered much of
117 the signal and was strongly aligned with Brock probability, indicating that the model could exploit

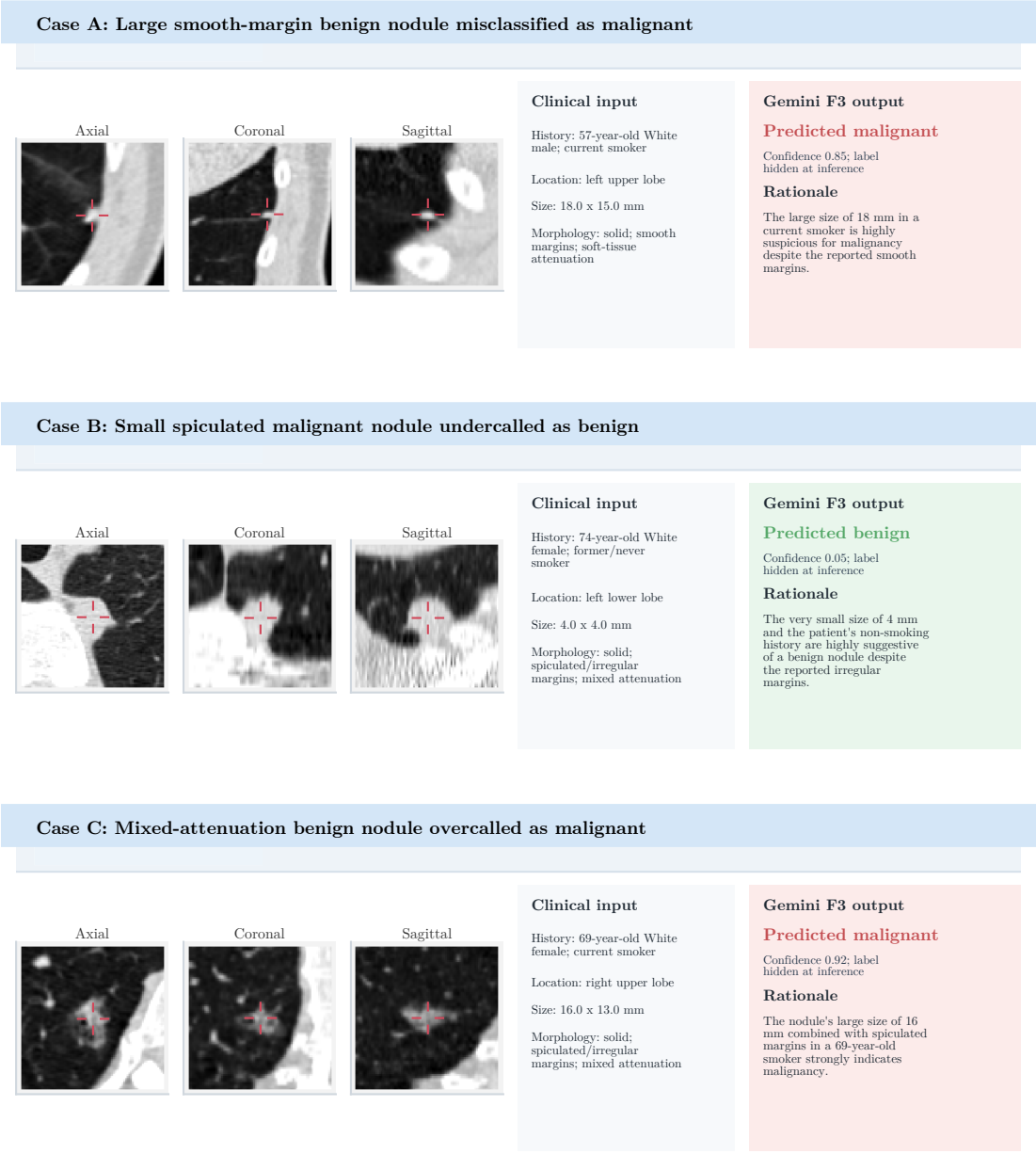


Figure 4: Representative discordant Gemini F3 cases on LUNA25. Each panel shows orthogonal CT views, supplied clinical input, emitted confidence and rationale, and the reference label for the F3 condition.

tabular/radiologic descriptors without CT images.

The cross-family audit supports the same interpretation beyond Gemini video prompting. With identical three-slice montage input, the Gemini 3 Flash bridge, Claude Opus 4.8, and Gemini 3.1 Pro were near chance without metadata but rose to AUC 0.72–0.73 when metadata were added; their text-only controls recovered most of this gain. GPT-5.5 metadata-assisted and text-only rows

showed the same approximate range, but its image-only row was retained only as an interface-provenance check because image delivery failed in the tested interface. For Gemini 3 Flash, the gap between native-video Z2 and montage Z2 despite matched nominal frame positions suggests that frame selection alone does not define the visual input: native video ingestion and single-image montage presentation can allocate visual detail differently. The key result is the convergence of metadata-assisted video, metadata-assisted montage, and text-only scores near AUC 0.72–0.73. This was not intended as a vendor leaderboard, because image/video ingestion differs across model families. Its value is narrower but clinically important: metadata reliance can masquerade as multimodal diagnostic performance across more than one frontier VLM family.

Task-specific supervised 3-D DL remained stronger. The comparison favored the VLM because Gemini F3 received CT video, 20 examples, and structured descriptors, whereas DL baselines used CT volume only. Even so, Gemini F3 was below six of seven DL baselines, had lower PR-AUC and net benefit than STU-Net, and was poorly calibrated. On LNDb, where comparable structured metadata were unavailable, visual-only Gemini remained below all DL baselines, reinforcing that metadata-assisted internal performance should not be generalized as visual CT competence.

Several limitations should be considered. LUNA25 labels combine pathology and longitudinal follow-up, and the deep ablation used one primary VLM family on one internal public cohort. The cross-family audit used a standardized three-frame montage rather than full 3-D CT or vendor-native video; even when sampled frame positions were matched to the Gemini video probe, montage and video inputs may differ in resizing, token allocation, and modality-specific preprocessing. GPT-5.5 image-only behavior suggested failed image delivery through that interface. Non-F3 VLM conditions were single runs, and calibration, subgroup, decision-curve, rationale, and weak-adaptation analyses were exploratory. We did not run metadata-permuted VLM inputs because deliberately mismatching descriptors and CTs is not a clinically meaningful diagnostic request. Self-reported race was included as an available NLST descriptor in Z3/F3 text; race-excluded tabular sensitivity analyses changed AUC minimally, but no powered VLM fairness audit was performed.

In conclusion, VLM evaluation for medical imaging should include metadata controls when prompts contain structured clinical or radiologic descriptors. For CT lung nodule malignancy prediction, the tested VLMs did not match supervised 3-D DL and should not be used as standalone diagnostic models. A more plausible near-term role is as an interface or reporting layer around

validated imaging models, tested prospectively with calibration and clinical workflow endpoints.

Data and code availability

LUNA25 is publicly available at <https://luna25.grand-challenge.org/>. The LNDb ROI external-validation labels and the 814-row 10:1 evaluation sheet used here are derived from the public LNDb cohort; raw CT data are not redistributed and should be obtained from LUNA25 and LNDb under their respective terms. Per-sample VLM and DL predictions (JSONL/CSV), cross-family audit outputs, the row-level LNDb external-validation CSV, the Brock prediction file, prompts, exemplar identifiers, and all code (CT-to-video conversion, prompt construction, inference orchestration, cross-family shard merging and analysis, the Brock implementation, and the statistical, calibration, decision-curve, and figure-generation scripts, with pinned package versions) are available in an anonymized review repository and will be made public under the Apache 2.0 license on acceptance. *[Repository link withheld for double-blind review; provided to editors and added on acceptance.]*

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Competing interests

The authors declare no competing interests.

Generative AI statement

During preparation of this work, the authors used Anthropic Claude and OpenAI GPT-family models for language polishing, consistency checks, and non-substantive organizational suggestions after all analyses were completed. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No AI tool generated new data, executed analyses, or created numerical results.

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